The Green Capital Gap

A deal-level analysis of cleantech venture funding in emerging economies, 2000–2012

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Abstract

Emerging-market green entrepreneurship is almost absent from the open venture record, and what absence there is turns out not to be about green. Across 41,105 funding rounds from the Crunchbase 2013 Snapshot, classified by a three-tier green classifier with inter-tier agreement of $\kappa = 0.850$, an emerging-market company already indexed in the record is **half as likely to be green as an advanced-economy one**: 2.60% against 5.43%, a risk ratio of 0.479 [0.343, 0.669]. That gap survives every green definition, every bloc scheme, and a permutation test in which **zero of 1,000** label shuffles reach the observed value.

The gap is **not green-specific**. Across 23 sectors, emerging-market representation falls roughly in proportion to capital intensity, with an elasticity of -1.19 [-1.64, -0.75] explaining 0.60 of the cross-sector variance; biotech sits further below proportionate representation than cleantech does. Cleantech's residual from that fit is **positive** at $1.92\times$ [$0.75\times$, $4.91\times$], so a green-specific penalty is ruled out.

Three of the five hypotheses could not be resolved, each for a different and separately verifiable reason, and together with three further measurement failures they give **six independent routes to one conclusion: the capital-regime question is not answerable from open data.** That is the paper's finding, not its limitation.

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1. Introduction

Green entrepreneurship is the private-sector channel of the climate transition. Public money sets prices and de-risks; firms build the plant. Where those firms raise capital, on what terms, and whether they survive the step from prototype to deployment determines how fast the transition moves where it matters most.

The money is not where the need is. Climate Policy Initiative puts tracked global climate finance at roughly US$1.9 trillion in 2023, with advanced economies and China accounting for around 80% of flows since 2019; emerging markets excluding China captured roughly 14–17%, and least-developed countries 2–3% (Climate Policy Initiative 2025). The same source puts the emerging markets grouping at over half the world's population and 23% of global GDP. Mitigation finance needs to reach roughly US$7.8 trillion a year over 2025–2030. The mismatch between need and flow is the motivating fact of this paper.

Evidence is missing at the level the mismatch is created: individual deals, with dates, amounts and instruments. Aggregate flow accounting cannot say whether an emerging-market green venture raises later, smaller, or on different instruments than its advanced-economy counterpart, because it does not observe ventures, and deal-level data that could say so is almost entirely paywalled. This paper asks what the open record can support, using the Crunchbase 2013 Snapshot (Crunchbase 2013) joined to World Bank and Our World in Data enrichment layers.

It does not succeed in measuring that regime, and the central claim is that doing so with open data is not possible — a result rather than a shortfall. Five pre-registered hypotheses were specified before the data was examined. Two returned usable answers within advanced economies; three could not be resolved, each for a different and independently verifiable reason. Three further measurement failures arise from the record's own structure. Six routes, one conclusion.

Two findings survive. Conditional on a company already being indexed, an emerging-market company is half as likely to be classified green as an advanced-economy one. That gap is real and robust, and it is **not** green-specific: it is a capital-intensity gap that green sits inside, and sectors with no environmental character at all sit further below proportionate representation than cleantech does.

2. Background and related work

The benchmark for what venture capital does to clean technology is the post-mortem on Cleantech 1.0. Gaddy et al. find that venture firms committed over US$25 billion to cleantech startups between 2006 and 2011 and recovered less than half of it, and that more than 90% of cleantech A-rounds made after 2007 failed to return the capital invested (Gaddy et al. 2017). The losses concentrated in materials, chemistries and hardware firms that never reached manufacturing scale; software was spared. Their conclusion is structural rather than cyclical: clean energy needs more capital over longer horizons than a three-to five-year fund life permits, so the venture model is mismatched to the asset.

That sets the prior here. If the venture model fitted cleantech badly in deep, liquid, dollar-denominated US markets, it should fit worse in thinner ones. Emerging-market green ventures earn in local currency and raise in hard currency, and hedging is expensive or unavailable, so investors demand a premium that compounds the capital-intensity problem. The missing middle — too large for grants, too unproven for banks, too slow for venture — bites hardest at first-of-a-kind commercial deployment, which is where these ventures sit.

The emerging-economy literature adds institutional quality as the moderator. Where regulatory enforcement is weak, the policy pull that creates green markets is absent and green ventures compete on price against incumbents with unpriced externalities. Porter and van der Linde's

argument that well-designed regulation can make environmental performance and competitiveness complementary (Porter and Linde 1995) is conditional on enforcement; Khanna and Palepu's institutional voids describe what happens when the intermediating institutions that argument assumes are missing (Khanna and Palepu 1997). The recurring empirical finding is that green capital allocates toward depth of financial markets and ease of exit rather than toward emissions, vulnerability or energy poverty.

Current trackers disagree about the present-day level. Net Zero Insights records US$23.5 billion of climate tech equity in H1 2025, one of the weakest half-years since 2020, with non-dilutive capital rising as equity shrinks and graduation rates from seed to Series A falling (Net Zero Insights 2025). Sightline Climate puts H1 2025 funding down roughly 19% year-on-year, with Series B down 29% (Sightline Climate 2025). Those estimates are not reconcilable without knowing each provider's taxonomy and instrument coverage. The disagreement is itself evidence for this paper's argument: the measurement of green entrepreneurship is not settled, and a classifier's construction determines what it counts. That is why the classifier in §4.1 is built in separable tiers and reported with its agreement statistic.

The gap this paper fills is not another estimate of the level, but a direct test of whether the open record can answer the structural questions the literature poses, and a demonstration of the specific points at which it cannot.

3. Data

The primary source is the Crunchbase 2013 Snapshot, distributed under CC BY 4.0 (Crunchbase 2013). Two Kaggle mirrors are used and joined. The relational dump carries deal-level rows — date, amount, round type and round code — but only a coarse single-value `category_code`. The flat export carries a pipe-delimited `category_list` rich enough to support a green taxonomy, but no deal dates. Joining them on a normalised permalink yields deal-level rows carrying rich green labels, which is what makes the classifier in §4.1 possible. The join succeeds for 97.3% of companies and 96.8% of deals.

The two mirrors are **different vintages**, which the project documentation originally assumed they were not. The relational dump's last funding date is 2013-12-12; the flat file extends to 2015-01-01 and is an early-2015 export. 2,186 companies appear in the flat file with funding the relational dump does not record, implying under-recording of about 6.5% of deals. The flat file is therefore used for its taxonomy only and never as a funding source. The comparison is in Appendix H.

`funding_rounds.csv` holds 52,928 rounds in total. Applying the 2000-01-01 to 2012-12-31 window and the cleaning rules leaves 41,105 deals, a retention of 77.7%; 11,823 rows are dropped and every drop reason is itemised in the attrition ledger (Appendix E), which is

complete by construction because each filter is applied through a ledger object that records its own before and after counts. Two cleaning decisions are worth naming here. 6,000 rows encode an undisclosed amount as a literal `0.0`; these are nulled rather than treated as zero-dollar rounds, after a sweep confirmed no other sentinel value was in use. 362 rows carry a currency that cannot be converted and are excluded from dollar series but retained in deal counts.

Country attribution is the other substantial cleaning step, and it is measured at two scales that should not be confused. At company scale, 108,563 of the 196,553 companies in the snapshot carry no country code; recovery from the flat mirror's country field and from country-code top-level domains fixes 5,743 of them, a recovery rate of 5.3%. Most of those companies never raised a recorded round, so the effect on the analysis is far smaller: at deal scale, 354 of the 41,105 in-window deals gain a country this way, and 1,526 remain unattributable — 1,520 with no code at all and 6 carrying a code that maps to no country. 23 vanity top-level domains — `.io`, `.co`, `.me`, `.tv` and similar — are refused by name with a reason each, because they signal branding rather than domicile (Appendix B). Unattributable deals are reported as their own bloc rather than dropped or assigned, and §5.2 reports an upper bound in which all of them are counted as emerging-market.

Enrichment comes from the World Bank and Our World in Data (World Bank 2025; Our World in Data 2025): historical income classifications for the time-varying bloc assignment, GDP, population, electricity access, renewable share, and the US GDP deflator. The Indian startup funding scrape (Sudalai Rajkumar 2020) supports the continuity probe in §7 and is never pooled.

One expected bias does not appear. If emerging-market companies were more thinly described, the classifier would under-detect them for a clerical reason. They are not: across 1,019 emerging-market companies the mean tag count is 2.63 against 2.24 for advanced economies (Mann–Whitney $p = 7.3\times10^{-18}$) — the opposite direction to the one that would explain the result away. The record's bias is in which companies it indexes, not in how well it describes them.

4. Methodology

4.1 The green classifier

Green status is assigned by a three-tier classifier whose every pattern lives in `config/green_lexicon.yaml`, not in code. Tier A is the snapshot's own `category_code` flag. Tier B is a case-insensitive lexicon of 135 regular expressions over the concatenated taxonomy, tag and description fields, mapped to 11 sub-domains. Tier C is an exclusion list that vetoes Tier B hits.

The headline definition is `green = TierA OR (TierB AND NOT TierC)`. Tier C never vetoes Tier A: a company the snapshot itself flags as cleantech is not removed by a lexicon exclusion. Agreement between the two independent tiers is $\kappa = 0.850$ on funded companies carrying a taxonomy.

Two design choices prevent whole classes of false positive. Bare tokens are not triggers — only compounds — because *green*, *carbon*, *waste*, *organic* and *clean* are all common in unrelated senses, and admitting them to subtract the errors later is a losing game. A fossil-fuel veto was added after inspection showed Crunchbase's `Energy` bucket mixes renewables with oil, gas and coal; without it an upstream petroleum company classified as green. 312 companies are vetoed in total, each with a recorded pattern and reason, in Appendix A.

4.2 Bloc assignment

Each deal takes the bloc its country held **in the year the deal closed**, derived from the World Bank's historical income classification. A deal in Poland in 2004 is not a deal in an advanced economy; a deal in Poland in 2013 is. China is reported separately throughout, mirroring Climate Policy Initiative's practice, and least developed countries are separated on the UN list with graduation years applied per year.

Country attribution required repair, and the allow-list governing it is the methodological point. Several country-code domains belong to emerging or least developed countries and are sold worldwide as vanity domains, so a naive mapping would have manufactured 127 Tuvaluan and 18 São Toméan *companies* out of domain marketing — 651 fabricated company attributions in all — in a project whose finding is that such ventures are absent. 29 domains are accepted and 23 refused by name, each with a reason, in Appendix B.

4.3 Window, deflation and censoring

Amounts are converted to constant 2015 USD with the US GDP deflator, with the nominal column retained alongside. Undisclosed amounts are encoded in the source as a literal zero rather than null, which would have dragged every median toward zero while reporting complete coverage; 6,000 rows are nulled, never imputed.

The window closes at 2012-12-31. The usual argument for truncating is recording lag; here that is secondary. The primary argument is differential coverage growth. Crunchbase's coverage expanded over the period and expanded faster for emerging markets than for the United States, which saturated early, so the final year is the most coverage-contaminated year for precisely the quantity the headline measures. Figure M1 shows the boundary measured rather than assumed.

4.4 Statistical approach

Proportions carry Wilson score intervals rather than normal approximations, which misbehave on small proportions. Deal-size comparisons use rank-based tests because the distributions are heavy-tailed. Bootstrap intervals use 10,000 resamples from a seeded generator. A minimum cell size of 30 was fixed in configuration before any hypothesis was tested, and no significance test is reported on a cell below it.

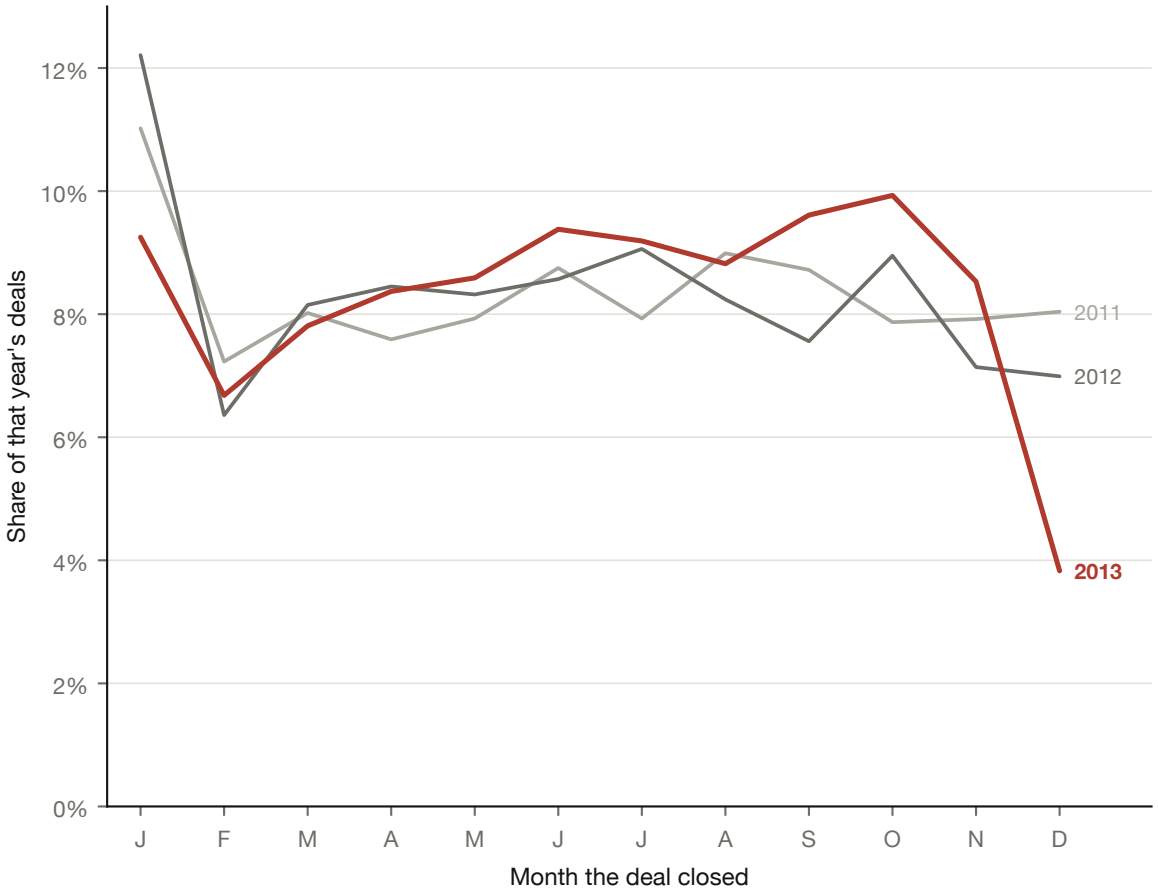


Figure M1. Only December 2013 is censored, which is why the window closes at 2012. Monthly deals as a share of their own year, 2011–2013. Expressing each month as a share of its own year removes the level difference between years so the shapes are comparable. December 2013 holds 3.8% of its year against a 7.5% baseline; the snapshot was taken mid-December, so only that month is truncated. November 2013 is above both earlier years (729 / 712 / 944), so coverage was not decaying. The disclosed-amount rate is flat across these years (88.99 / 88.61 / 88.44%), so dollars are not differentially censored either. The window still closes at 2012-12-31, on the coverage-growth argument rather than this one (§4).

Effect sizes accompany every p-value, because with tens of thousands of deals almost any difference reaches significance: the instrument-mix χ^2 in §6.3 is significant at $p = 1.2\times10^{-7}$ on a Cramér's V of 0.027, which is why the risk ratio is quoted instead. 28 tests carry Benjamini-Hochberg adjusted p-values in Appendix D, of which 9 survive correction.

The capital-intensity model in §5.3 regresses log emerging-market representation on log median advanced-economy deal size across 23 sectors. The proxy is built from advanced-economy deals only, so it cannot be contaminated by the emerging-market composition it is used to explain. Figure M2 shows the fit.

4.5 Specifications fixed in advance

Four judgement calls that move a headline number were fixed in configuration, with reasoning recorded, before the relevant result was computed: whether a bare `Energy` tag admits a company as green, whether angel rounds count as funnel entry, that H3's grant arm is descriptive only below the minimum cell size, and that H4's extended window is permitted because it conditions

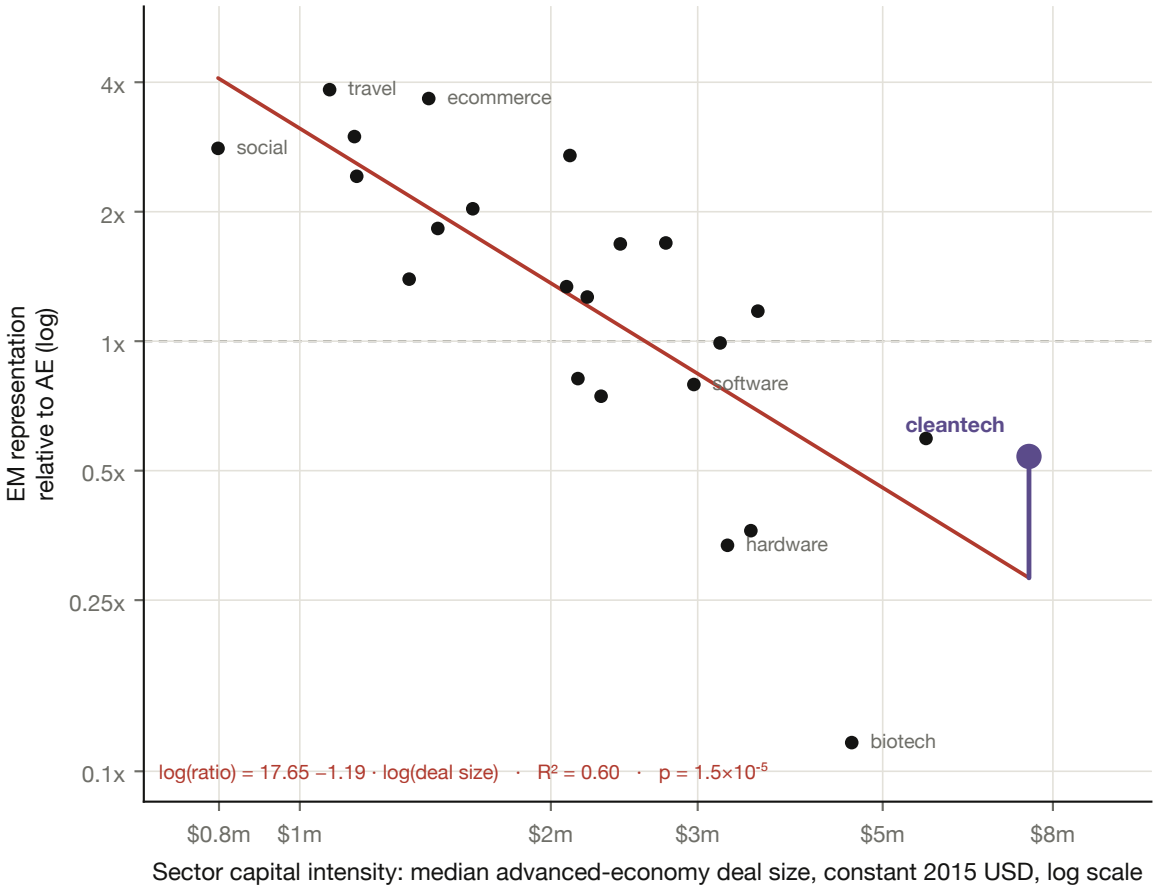


Figure M2. Emerging-market representation falls roughly in proportion to capital intensity. 23 sectors, 2000–2012. Ordinary least squares on logs; the line is the model's fitted values. Elasticity −1.19 [−1.64, −0.75], $R^2 = 0.60$, $p = 1.5\times10^{-5}$. Cleantech sits 1.92× above the fit [0.75×, 4.91×]. The proxy uses advanced-economy deals only. The secondary proxy, median cumulative funding per company, gives an elasticity of −0.96 with R^2 0.69.

on sample entry. Each is run both ways in Appendix C. Where a pre-commitment's assumption proved wrong — H4 assumed a direction when the binding failure was power — it is recorded as void rather than quietly satisfied.

4.6 Reproducibility

This report was produced by 4 authors — Kaizer Dewaswala (1RV23IS058); Md Asif Ali (1RV23IS069); Shaswat Kumar (1RV23IS110); Vansh Bharadwaj (1RV23IS134) — under the guidance of Prof. ChandraKumar R at the Department of Information Science & Engineering, RV College of Engineering. That list lives in `report/metadata.yaml`, read from there by the title page, this statement and the README.

`make all` reproduces every number from a clean checkout. All 12 source files carry recorded SHA256 checksums, verified each run, so a changed upstream file fails the build rather than silently changing a result. Every statistic here is injected from `outputs/findings.json` and the renderer fails on an unresolved placeholder, so no number in this report was typed by hand — including the numbers in the figure captions. Commit 87c48f9, Python 3.11.9, seed 20251001.

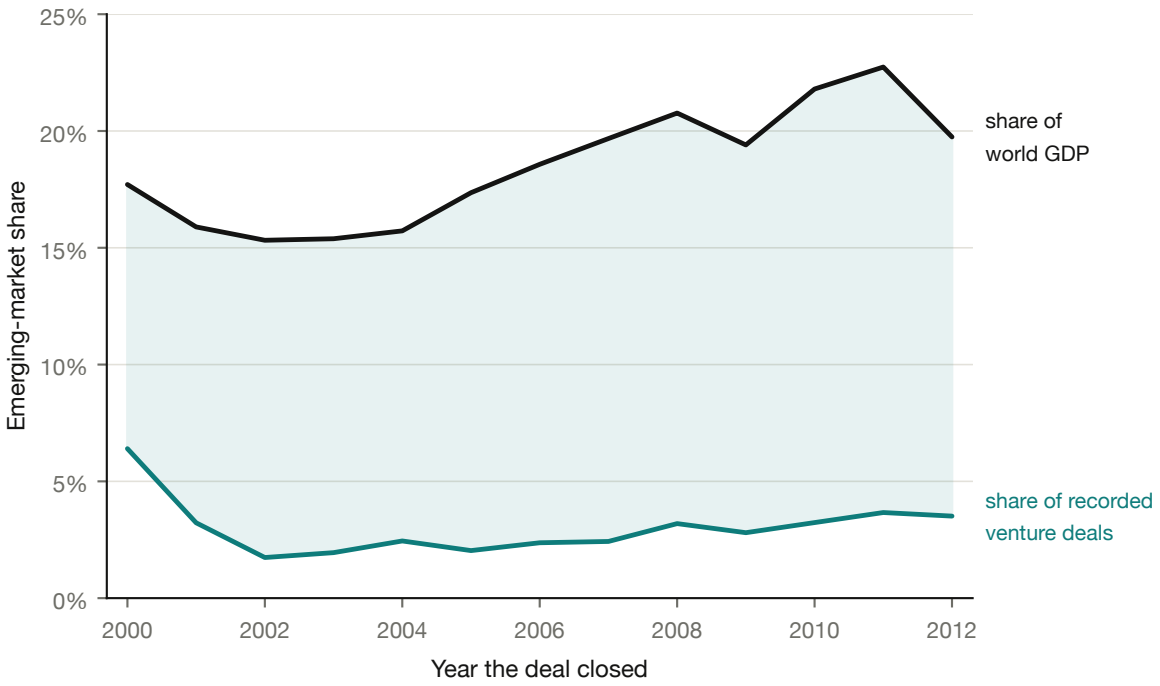


Figure 1. Emerging markets hold a fifth of world GDP and a fiftieth of the venture record. Emerging-market share of world GDP against its share of recorded venture deals, 2000–2012. GDP shares are computed over the same country set the bloc panel covers, so the two series are comparable by construction. Across the window emerging markets hold 15.3–22.7% of world GDP against 1.7–6.4% of recorded venture deals, an under-representation of 2.8–8.8×. This is the denominator check: it establishes that the conditional green-rate gap is not this coverage gap restated.

5. The double gap

5.1 Emerging markets are a fiftieth of the record and a fifth of world GDP

Over 2000–2012 emerging markets held 15.3–22.7% of world GDP and 1.7–6.4% of recorded venture deals, an under-representation of 2.8–8.8× (Figure 1). This is the coverage gap, and it is the reason the paper's headline result is measured *within* the dataset rather than against the world.

5.2 Conditional on being in the record, an emerging-market company is half as likely to be green

1,309 emerging-market deals carry 2.60% green [1.86, 3.61], against 5.43% [5.20, 5.66] across 37,896 advanced-economy deals. The difference is −2.83 percentage points [−3.72, −1.94]; the risk ratio is 0.479 [0.343, 0.669], z-test $p = 7.5\times10^{-6}$, Fisher's exact $p = 1.3\times10^{-6}$ (Figure 2).

Both numerator and denominator are drawn from companies Crunchbase chose to index, so the result does not depend on the coverage gap in §5.1. Counting every deal with no recorded country as emerging-market — the most generous possible treatment — gives 3.46% and a risk ratio of 0.637, $p = 6.3\times10^{-6}$.

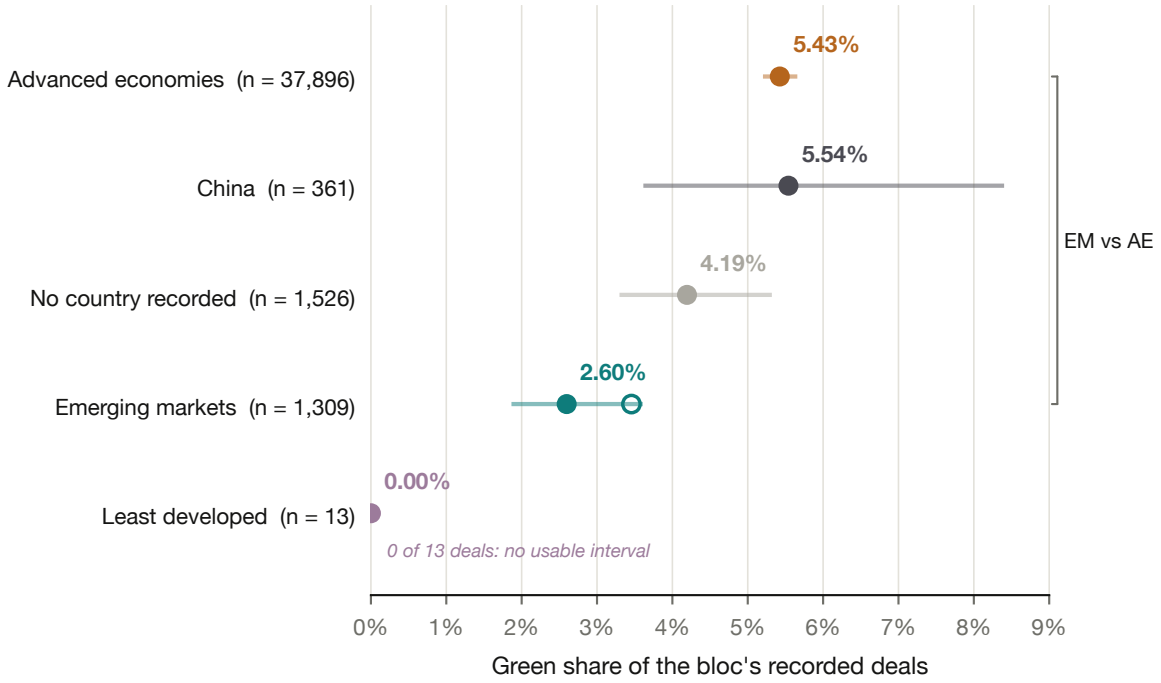


Figure 2. In the record, an emerging-market company is half as likely to be green. Green share of recorded venture deals by bloc, 2000–2012. Crunchbase 2013 Snapshot (CC BY 4.0); authors' calculations. n = 41,105 deals, 2,174 green; bloc rows sum to the total. Intervals are Wilson score. The bracket marks the headline comparison: 2.60% against 5.43%, risk ratio 0.479 [0.343, 0.669], $p = 7.5\times10^{-6}$. Counting every unattributable deal as emerging-market, the most generous reading available, still leaves a rate of 3.46% against 5.43% and a risk ratio of 0.637. China is not distinguishable from advanced economies (risk ratio 1.02, p = 0.924) and is reported separately throughout. Headline is raw classifier output; bias-corrected risk ratio 0.43 [0.26, 0.64], curated 0.40, Tier A only 0.54 (Appendix C).

China, reported separately throughout, sits with the advanced economies rather than with emerging markets. Figure 3 shows the two rates by year, with every emerging-market cell annotated by its count.

Verdict: Supported. Falsified if the emerging-market green rate were indistinguishable from the advanced-economy rate, or if the gap vanished under the upper bound.

5.3 The gap is a capital-intensity gap

Emerging-market venture capital in this period concentrated in capital-light consumer internet and avoided capital-intensive sectors as a class. Regressing log emerging-market representation on log median advanced-economy deal size across 23 sectors gives an elasticity of −1.19 [−1.64, −0.75], $p = 1.5\times10^{-5}$, $R^2 = 0.60$ (Figure 4). Doubling a sector's capital intensity roughly halves its emerging-market representation.

Cleantech sits **above** the position that fit predicts, at 1.92× [0.75×, 4.91×] of its predicted representation. The interval includes 1, so outperformance is not established; what is established is that there is no green-specific penalty once capital intensity is accounted for. Biotech, by contrast, sits below its own prediction.

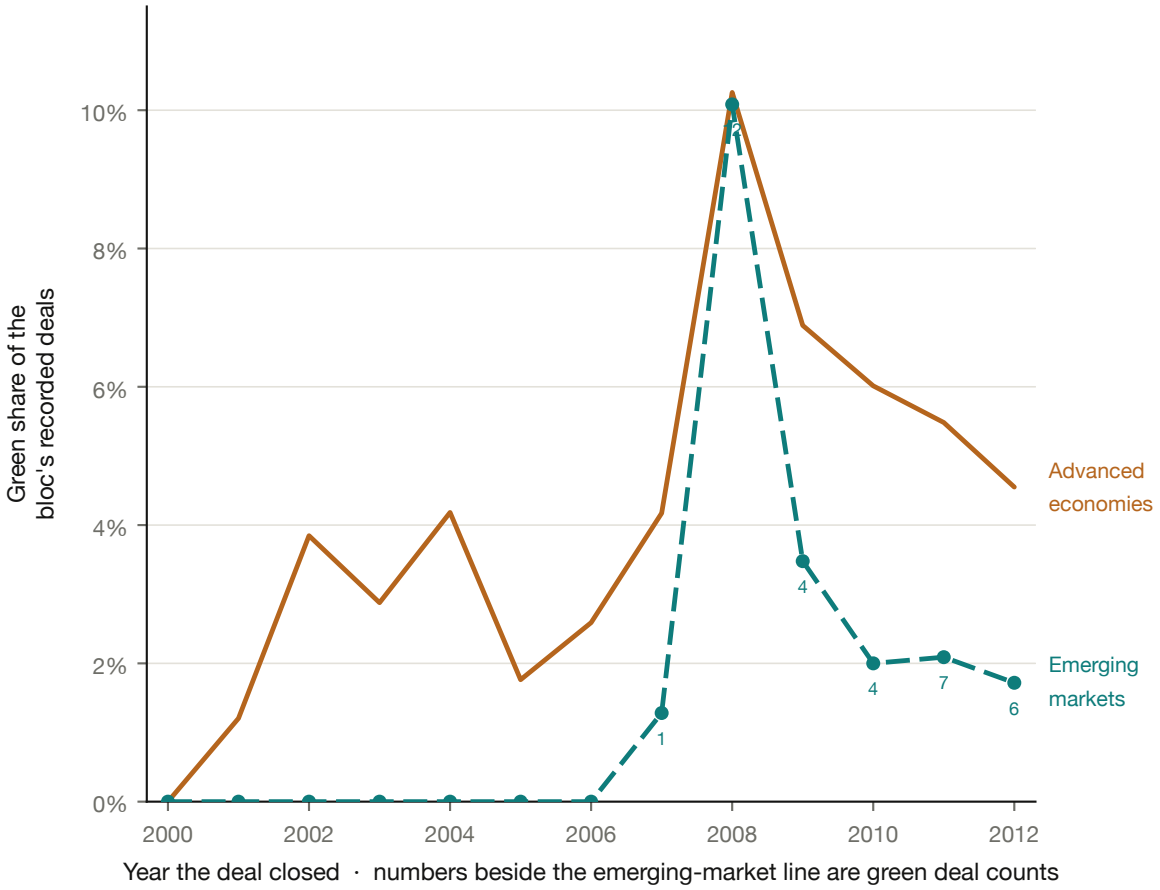


Figure 3. The two rates met in 2008, at the peak of Cleantech 1.0, then diverged. Green share of recorded deals by bloc and year, 2000–2012. No emerging-market year-cell reaches the minimum cell size of 30 green deals; the largest is 12, in 2008. In that year the two rates converge, and the gap opens only after the crash. No per-year test is reported and this figure is descriptive only; counts are printed beside each emerging-market point so the reader can see the precision available.

Green also took a larger share of venture dollars than of venture deals from 2005 onward (Figure 5), which is consistent with green being capital-intensive without testing the cross-sector elasticity above.

5.4 The taxonomy cannot say what kind of green

88.2% of emerging-market green deals carry no sub-domain more specific than “Clean Technology”, against 67.9% in advanced economies (Figure 6). The mechanism is direct: 23 of the 27 emerging-market green companies carry no other taxonomy tag at all. The planned sub-domain mix comparison was cut, because the measurement failure shown here makes it impossible.

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6. The advanced-economy benchmark, and what each null says about the record

Advanced economies are the one bloc dense enough to test the structural hypotheses, and the benchmark case for the capital-model mismatch Gaddy et al. identify. Each subsection closes by stating what its emerging-market arm shows about the record itself.

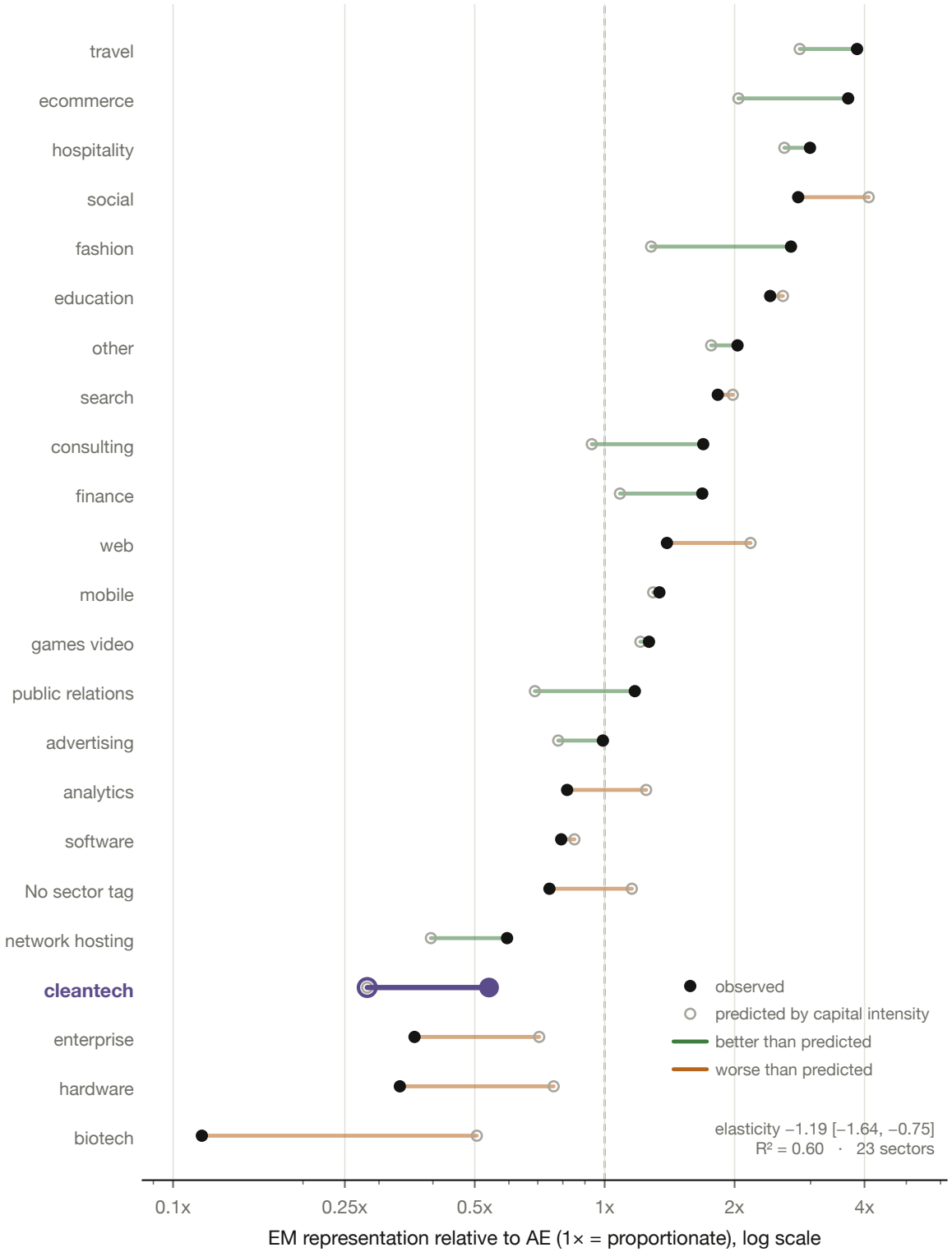


Figure 4. Capital intensity drives the pattern, and green beats its prediction. Emerging-market representation relative to advanced economies across 23 sectors, 2000–2012. Capital intensity is median advanced-economy deal size in constant 2015 USD, so the proxy cannot be contaminated by the emerging-market composition it explains. Segments run from predicted to observed; intervals are residual 95% CI. Cleantech sits 1.92× above its predicted position [0.75×, 4.91×]: capital intensity over-predicts the gap, and because the interval includes 1 the outperformance is not confirmed. Biotech is further under-represented than cleantech and also sits below its own prediction.

6.1 Green deal sizes fell far faster than non-green within the same stages

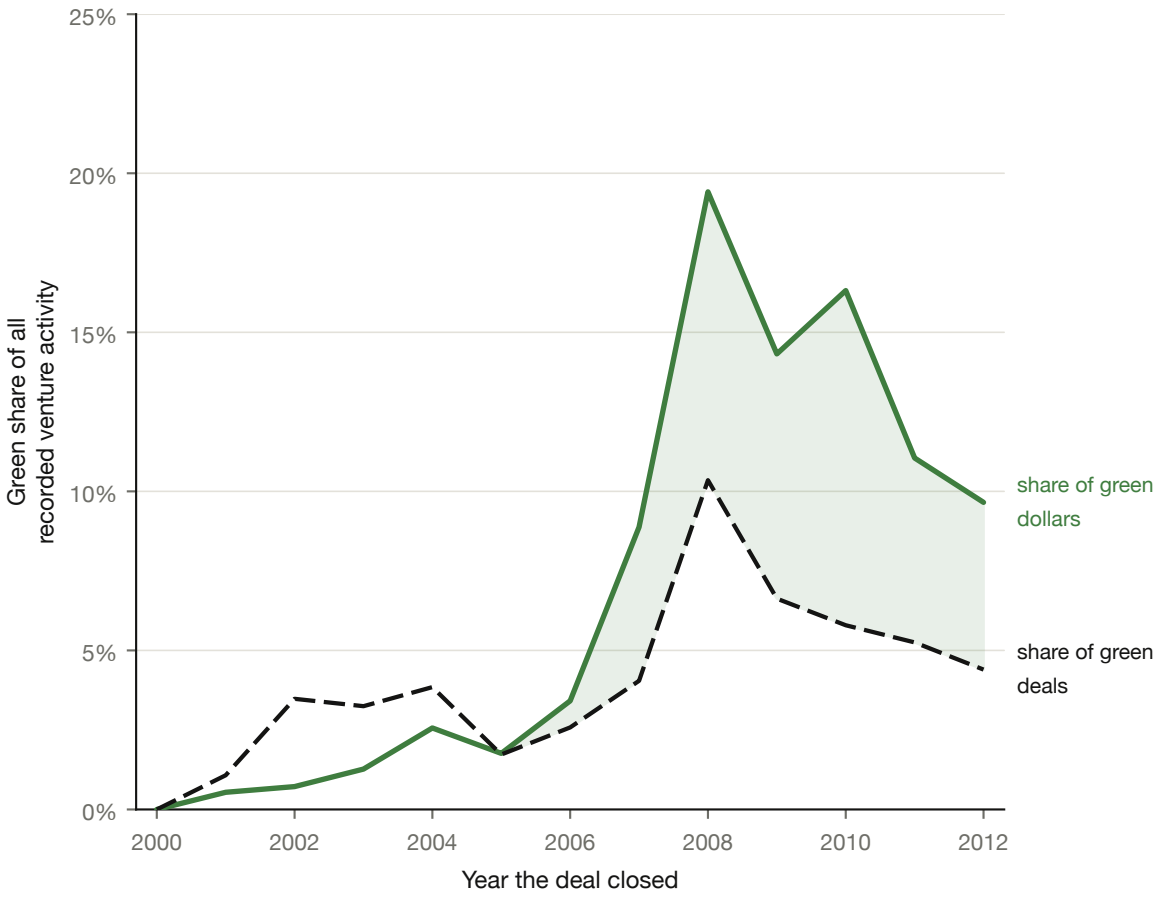


Figure 5. From 2005, green took a bigger share of venture dollars than of deals. Green as a share of all recorded deals and of all disclosed dollars, 2000–2012. Constant 2015 USD. A dollar share above the count share means the average green cheque exceeded the average cheque overall, which is consistent with green being capital-intensive; it does not test the cross-sector elasticity. The two series cross in 2005: before then a green deal was smaller than the average deal, and from 2005 on it was larger. The ordering reverses before 2005 on thin cells (2001 n=1, 2002 n=4, 2003 n=5, 2004 n=11).

Between 2007 and 2012, within named funding stages, green deal sizes fell −50.4% against −1.1% for non-green — a difference of −49.3 percentage points [−69.3, −22.4], bootstrapped over 10,000 draws (Figure 7).

The pooled median falls by about the same amount in both groups and settles nothing: it is composition-confounded for each, because the platform recorded far more small rounds as it matured. All 5 start-end pairs running to the close of the window are negative, spanning −69.9 to −21.5 percentage points, and the headline is their median, so the direction is robust to the start year. It is **not** robust to the end year: across 17 admissible pairs the difference is negative in 71%, and every positive pair stops in 2006 or 2009–11, before the decline has run its course. The claim the grid supports is a direction holding to the window close, with a magnitude depending on where the comparison starts.

This is a size pattern, not an investor motive. The rotation out of hardware Gaddy et al. document is cited as external evidence, not as something this dataset shows.

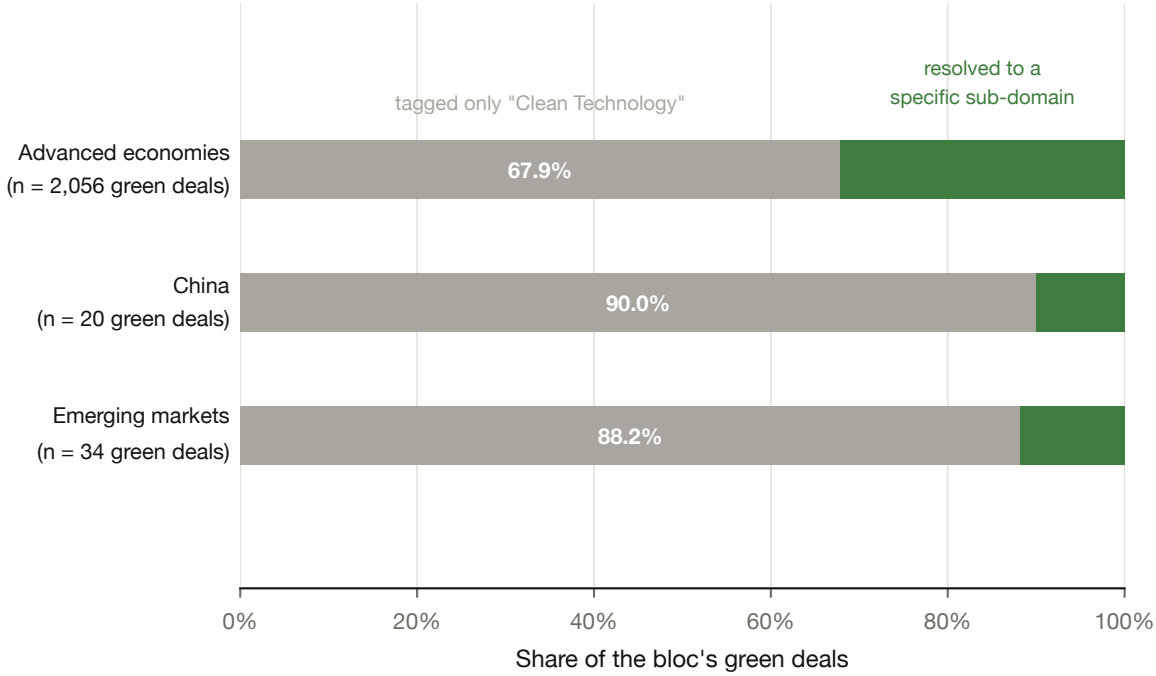


Figure 6. The taxonomy cannot say what kind of green an emerging-market company is. Share of green deals resolved to a specific sub-domain, by bloc, 2000–2012. 23 of the 27 emerging-market green companies carry no tag more specific than "Clean Technology", against 67.9% of advanced-economy green deals. The 11 specific sub-domains are solar, wind/hydro/geothermal, biofuels, storage, efficiency and smart grid, electric mobility, water, waste and circular, sustainable agriculture, green building, and environmental monitoring. This figure replaces the planned sub-domain mix comparison, which the measurement failure shown here makes impossible.

6.2 The crisis hit green and spared non-green, but the test does not identify it

Across 2008–09, advanced-economy green deals contracted −26.5% while non-green **grew** 13.5% (Figure 8).

The difference-in-differences designed to test this does not identify it. Green deal volume was already growing 0.254 log-points per year faster than non-green before the event window, [0.101, 0.408], p = 0.001. Parallel trends fail, so the design never recovered a causal effect and there is no null to report.

Verdict: Not testable — identifying assumption failed. The contraction figures above are descriptive.

What the emerging-market arm shows about the record: it does not exist. The largest emerging-market green year-cell holds 12 deals.

6.3 Green capital is less dilutive than non-green

Within advanced economies, 10.41% of green deals use non-dilutive instruments — grants, debt, post-IPO debt — against 7.26% of non-green deals. The risk ratio is 1.433 [1.256, 1.636], $p = 1.2\times10^{-7}$ (Figure 9).

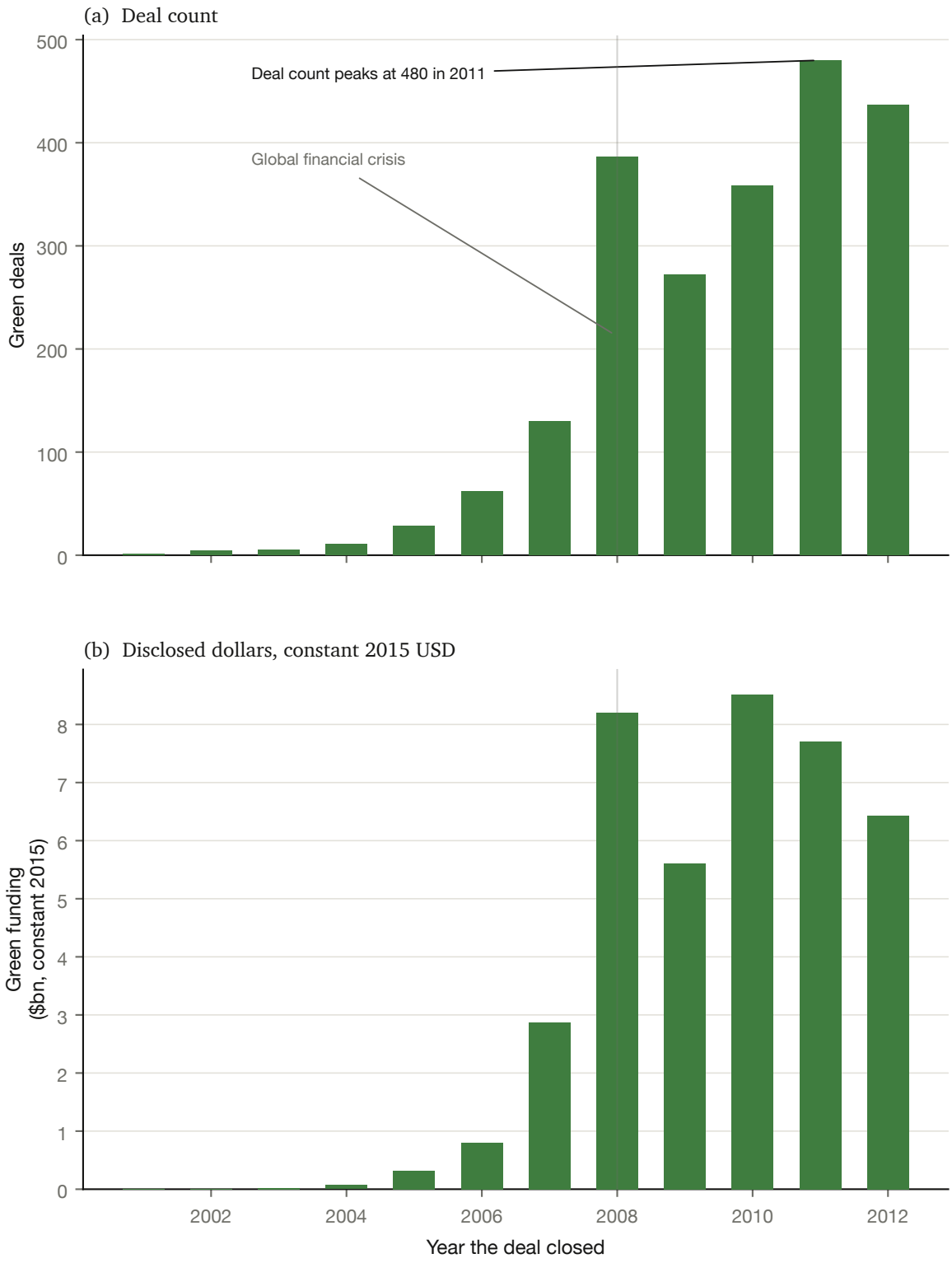


Figure 7. Green deal sizes fell far faster than non-green within the same stages. Green deals and disclosed green funding, all blocs, 2000–2012. Constant 2015 USD. n = 2,174 green deals, 1,994 with a disclosed amount. Counts and dollars are separate series because undisclosed-amount rates differ by instrument and bloc. Dollars peak a year before deals do. The cheque-size claim is the within-stage comparison: green −50.4% against non-green −1.1%, difference −49.3pp [−69.3, −22.4], bootstrapped over 10,000 draws on clean stage codes. The pooled median falls alike in both and is composition-confounded for each. Robust to the start year, sensitive to the end year (§6.1).

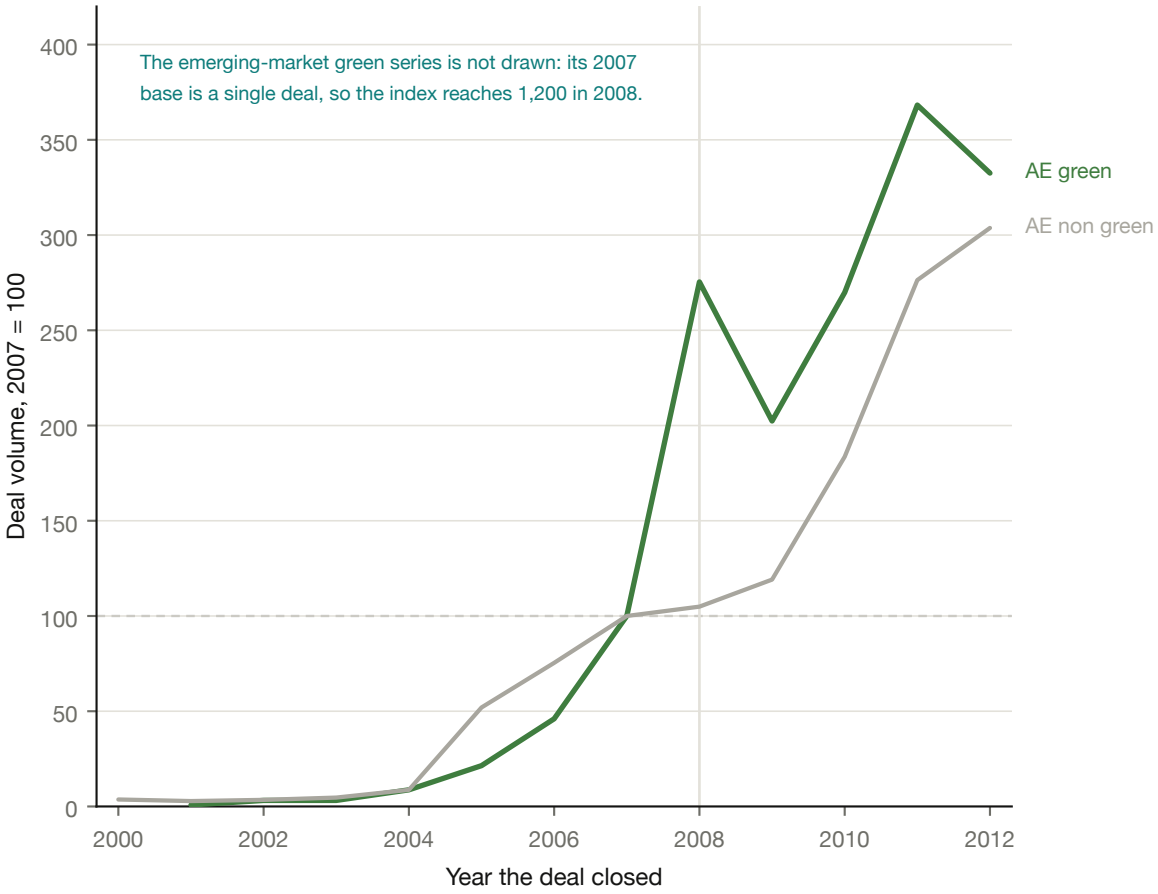


Figure 8. The crisis hit green venture funding and left non-green growing. Deal volume indexed to 2007, by bloc and green status, 2000–2012. Across 2008–09 advanced-economy green contracted −26.5% while advanced-economy non-green grew 13.5%. The emerging-market green series is not drawn: its 2007 base year holds 1 deal, so the index would reach 1,200 in 2008. No difference-in-differences coefficient is shown: green volume was already growing 0.254 log-points per year faster than non-green before the event window ($p = 0.001$), so parallel trends fail, these series are descriptive, and the design does not identify a causal effect.

The χ^2 on the full instrument mix is also significant, but its Cramér's V of 0.027 falls below the conventional threshold for a small effect: with tens of thousands of deals the test detects a negligible association, and the risk ratio on the quantity the hypothesis is about is the interpretable figure.

Verdict: Supported for the advanced-economy benchmark case.

What the emerging-market arm shows about the record: it is null by construction. The record codes infrastructure project equity as ordinary venture rounds, which understates non-dilutive share on both sides and more so for emerging markets, where independent power producers are 7 of 27 green companies. The direction of that bias was stated before the numbers were computed. On 34 deals the arm could not have settled the question either way.

6.4 The scale-up step is unobservable

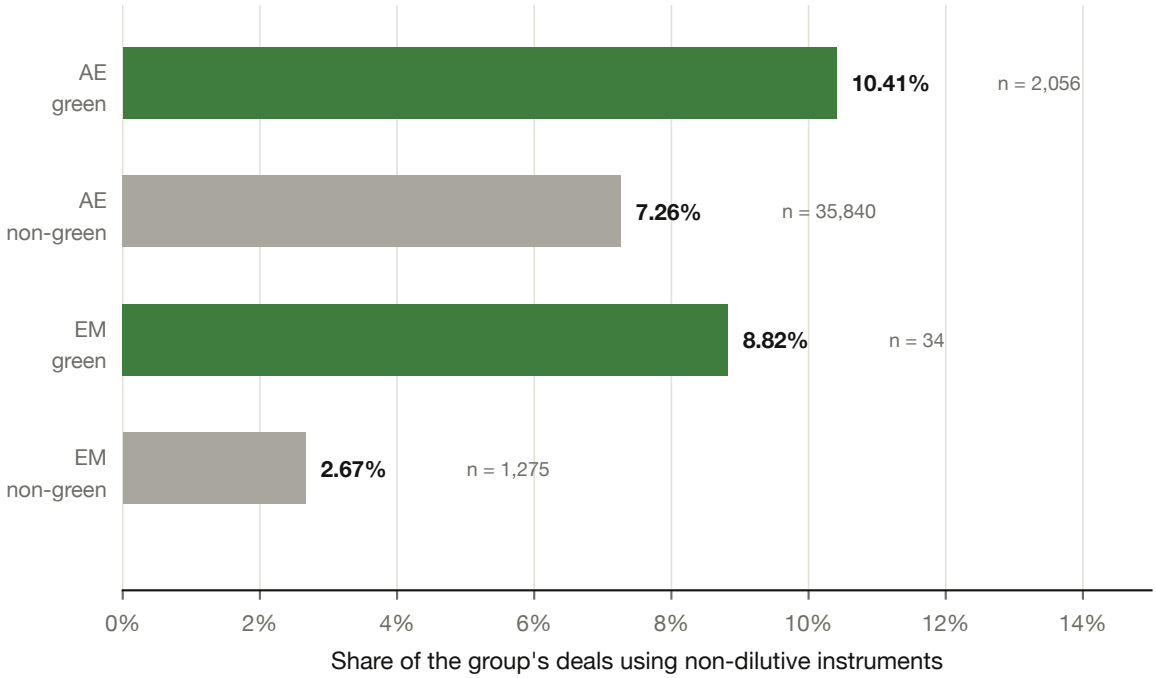


Figure 9. Green ventures lean harder on grants and debt than non-green ones do. Non-dilutive share of deals by bloc and green status, 2000–2012. Non-dilutive instruments are grants, debt financing and post-IPO debt. Within advanced economies, green uses one on 10.41% of deals against 7.26% for non-green: risk ratio 1.433 [1.256, 1.636], $p = 1.2\times10^{-7}$. The emerging-market arm rests on 34 deals, of which 3 are non-dilutive, and is reported as a pre-registered null rather than as an estimate. Project finance that is debt or concessional in substance but coded as an equity round understates the non-dilutive share on both sides, and more so for emerging markets, where utility-scale generation accounts for 7 of 27 green companies; that bias direction was stated before these numbers were computed.

Conditional on raising seed capital, advanced-economy green companies reached Series A with an odds ratio of 1.30 [0.65, 2.60] against non-green, $p = 0.458$. The interval spans 1 and identifies no direction.

The question is whether a wider window or a looser seed definition would resolve it. It would not. The advanced-economy green seed cohort holds 40 companies and the Series A to Series B pool — the specific missing-middle transition — holds 13. No model of that transition is reported, under any specification.

Verdict: Underpowered — cannot determine.

What the emerging-market arm shows about the record: there are 0 emerging-market green seed entrants across thirteen years. The scale-up step the missing-middle literature identifies as decisive cannot be observed at all.

6.5 Capital tracks market depth; whether it tracks need cannot be determined

Across 22 countries with enough green activity to measure, green funding per capita correlates with GDP per capita at a Spearman ρ of 0.674 [0.298, 0.866] (Figure 10). Green funding is extraordinarily concentrated: a Gini of 0.917, with the top five countries holding 89.9% of all green dollars across 42 countries.

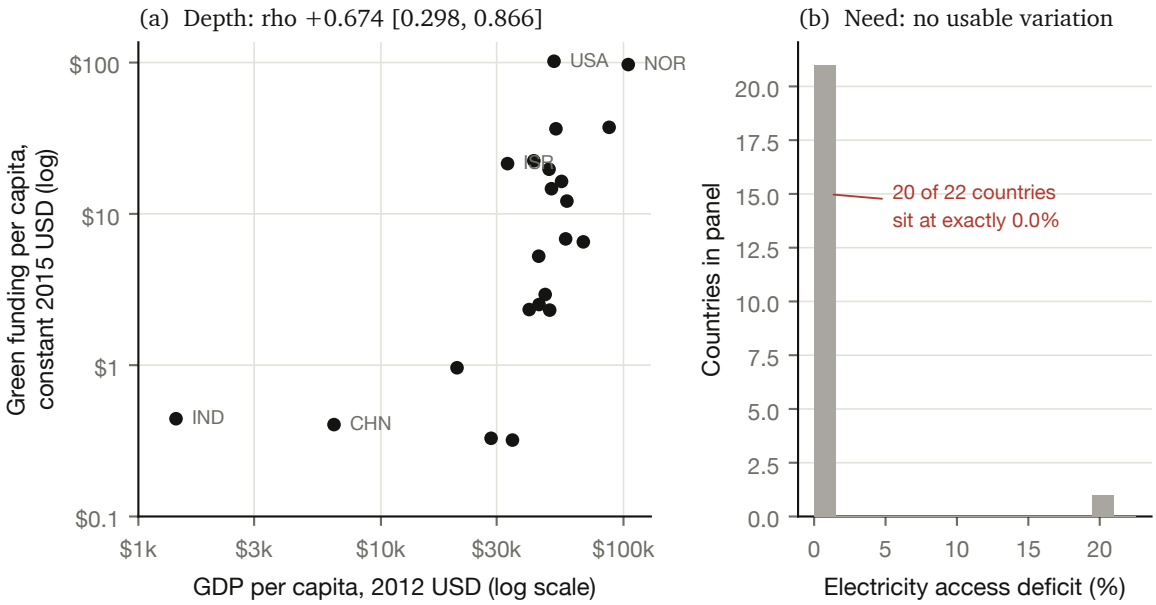


Figure 10. Capital tracks market depth; whether it tracks need cannot be measured here. Green funding intensity across 22 countries with at least three green deals, 2000–2012. Spearman rank correlation 0.674 [0.298, 0.866], with a bootstrap interval over 10,000 draws. The planned small-multiple of need indicators is not drawn; panel (b) shows why. A scatter against a regressor on which 20 of 22 countries record exactly zero would read as a relationship and is not one. Green funding is highly concentrated: Gini 0.917, top five countries hold 89.9% of green dollars.

The need limb cannot be tested. 20 of the 22 panel countries record an electricity-access deficit of exactly zero; the regressor is in substance a dummy for one country. A need regressor with no variance cannot identify anything, and a scatter against it would present a relationship that does not exist. An income-controlled coefficient on it appeared positive and significant, met the pre-registered falsification condition, and survived both leave-one-out and collinearity checks before a degenerate-variance check rejected it.

Verdict: Partially supported. The depth limb holds; the need limb is untestable here. The depth result would be falsified if the correlation reversed or its interval crossed zero; the need limb would become testable only on a panel with real variation in the need indicators.

What this shows about the record: 42 countries record any green deal at all and only 22 carry enough to enter the panel — and those are the countries that already have capital markets. The countries where need is highest are precisely the countries with too few green deals to enter the panel, which is not a missing variable but the same selection this paper has been describing throughout.

6.6 Six routes, one conclusion

Three hypotheses were unresolved, each for a different stated reason: H2′s identifying assumption failed, H4 had 13 companies at the transition it exists to measure, H5′s need regressor had no variance. Three further failures are properties of the record rather than of any hypothesis: 88.2% of emerging-market green deals carry no specific sub-domain; a pre-

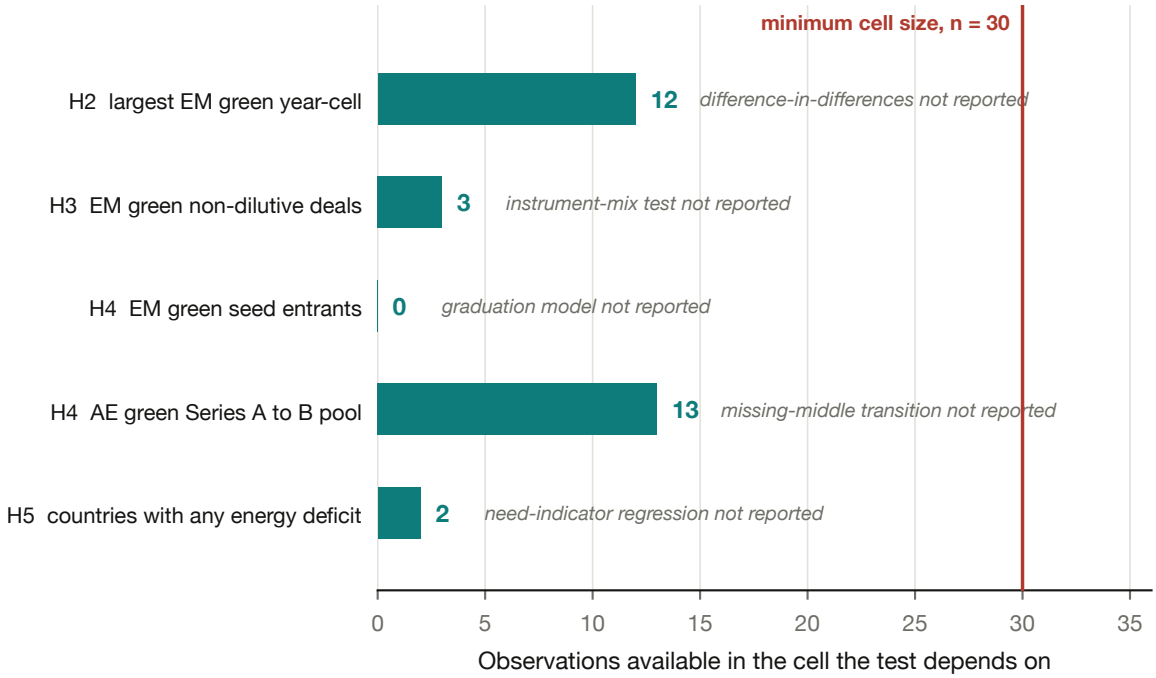


Figure 11. Every emerging-market test fails on the same wall, for different reasons. Observations available against the pre-registered minimum cell size, across H2–H5. The minimum cell size of 30 was fixed in config before any hypothesis was tested, and no bar clears it. These are five distinct failures rather than one repeated. The emerging-market green sample of 34 does itself clear the threshold; the non-dilutive cell the instrument test depends on holds 3. Three figures that would have plotted these quantities as estimates were cut rather than drawn.

declared venture-stage criterion failed because project finance is coded as venture rounds; and H3′s emerging-market arm is biased by that same coding in a direction stated in advance.

Figure 11 puts the first four on one scale, against the minimum cell size fixed in config before any hypothesis was tested. No cell clears it.

These are six independent routes to one conclusion. The open venture record cannot support the capital-regime analysis the field needs.

7. What the open record cannot answer

The capital-regime question is not answerable from open data. That is this paper's finding, and it is a specific claim rather than a call for more research. Five pre-registered hypotheses about how emerging-market green ventures raise capital were specified before the data was examined. Each failure has a named, separately verifiable cause, and no two causes are the same: an identifying assumption that the data falsifies, a transition cell holding 13 companies, a regressor with no variance, a taxonomy that resolves 88.2% of emerging-market green deals no further than "Clean Technology", and an instrument field that codes infrastructure project equity as an ordinary venture round. None of these would be fixed by a longer window or a looser classifier. They are properties of the data infrastructure.

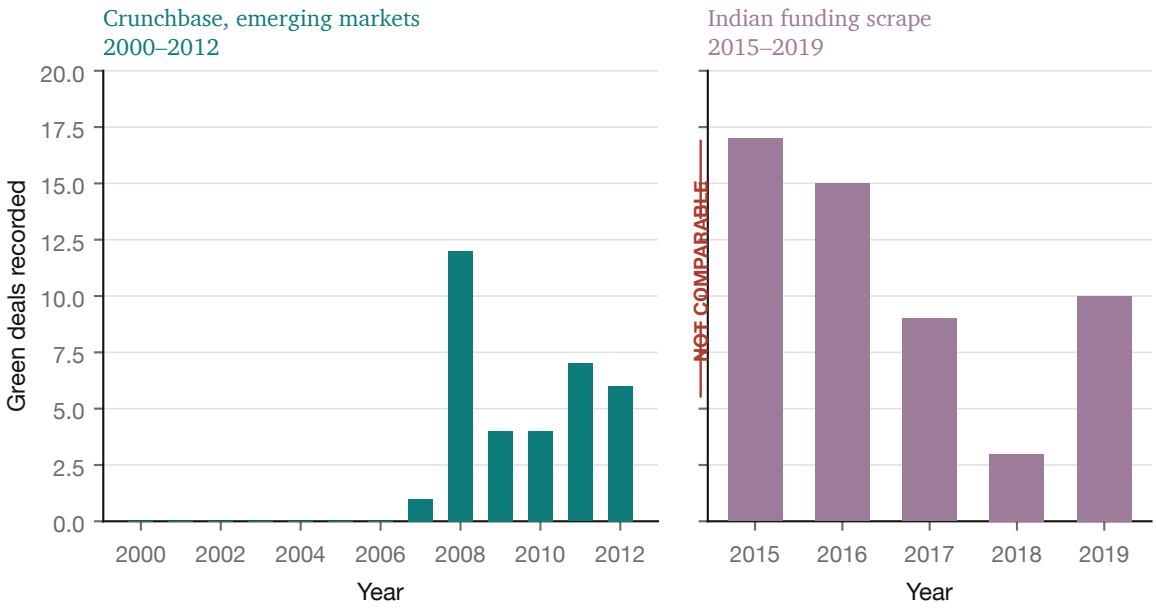


Figure 12. A later window in one large emerging market reproduces the same scarcity. Green deals per year. Two different instruments measuring two different populations; the panels share no axis. Left: Crunchbase 2013 Snapshot, emerging markets excluding China and LDCs, 34 green deals. Right: the Indian startup funding scrape, 3,036 dated deals of which 54 are green under the same lexicon. Never pooled: taxonomies and amount conventions are not compatible, and no category_code exists in the Indian data so Tier A cannot fire. 2020 is a partial year and is excluded.

The continuity probe asks whether the scarcity is an artefact of the window. It is not. The Indian startup funding scrape covers 2015–2020 with 3,036 dated deals, of which 54 are green under the same lexicon — 1.78%, against 2.60% for emerging markets in the Crunchbase window (Figure 12). The two are never pooled: the taxonomies are not compatible, the amount conventions differ, and no `category_code` exists in the Indian data so Tier A cannot fire. A later window in one large emerging market reproduces the same scarcity rather than escaping it.

Secondary evidence suggests two signatures persisted. Non-dilutive instruments were 7.2% of in-window deals here; Net Zero Insights reports non-dilutive capital rising as equity shrinks in 2025, and graduation rates from seed to Series A falling (Net Zero Insights 2025) — the same transition this paper could not observe at all. Climate Policy Initiative's 14–17% share for emerging markets excluding China (Climate Policy Initiative 2025) is far above the 1.7–6.4% deal share seen here, which is consistent with a large share of that capital moving through instruments a venture database does not index. Tracxn puts Indian climate tech at roughly US$12.8 billion cumulative across 1,583 funded companies since 2008 (Tracxn 2026), against the 54 green deals the open scrape records for 2015–2020. This section is suggestive and is not a test.

8. Discussion

The first implication is for the data infrastructure of the field, because that is where this paper bites. Green entrepreneurship in emerging economies is studied through aggregate flow

accounting because deal-level evidence is paywalled, and the open alternative cannot carry the weight. A public deal-level register with a stable green taxonomy and an honest instrument field would change what is answerable. The instrument field is the cheapest fix and the most consequential: a flag distinguishing project finance from ordinary venture equity would have made H3′s emerging-market arm testable and saved the venture-stage criterion this project had to abandon.

The second implication follows from the capital-intensity result. Emerging-market venture capital in this period avoided capital-intensive sectors as a class, and cleantech's position is explained by where it sits on that gradient rather than by anything about being green. Policy aimed at closing a green-specific aversion would be aimed at something the data does not show. What the data does show is a capital-structure problem: the sectors that need the most money per firm over the longest horizon are the ones thin markets serve worst. Concessional and blended capital bind at exactly that point, which is the mechanism the missing-middle literature names and which this result is consistent with.

For founders the implication is narrower and better evidenced within advanced economies, where the sample supports it: green ventures lean harder on grants and debt than non-green ones do, at 10.41% of deals against 7.26%, and green cheque sizes fell far faster than non-green through the Cleantech 1.0 unwind even within the same stages. Neither result can be extended to emerging markets from this data, and this paper does not extend them.

9. Limitations and threats to validity

Coverage bias is the largest threat and is not fully mitigable. Crunchbase 2013 under-covers non-US, non-English-language ecosystems, inflating the advanced-economy share of everything. Three things limit the damage: the headline is a comparison *within* the dataset, conditional on a company already being indexed, so it does not require the blocs to be covered equally; the coverage gap is measured rather than disclaimed (§5.1); and the taxonomy-richness check (§3) rules out the clerical version. What remains is selection into the record itself. If the emerging-market companies that reach Crunchbase are systematically less green than those that do not, the gap is overstated, and nothing in this data can exclude that.

The emerging-market green sample is 34 deals. That forbids essentially every disaggregation: by year, by sub-domain, by instrument, by country. The minimum cell size of 30 was fixed in config before any test was run, and no emerging-market cell clears it. Every emerging-market quantity in this paper is reported with its cell size printed, and no significance test is reported below that minimum.

The two-vintage join bounds how much of the record is missing rather than resolving it: roughly 6.5% of deals appear to be under-recorded in the relational dump. **Right-censoring**

is handled by truncating at 2012-12-31, argued from coverage growth rather than from the December artefact alone (§4.3, Figure M1). **Undisclosed amounts** mean deal counts and deal dollars are reported as separate series throughout; dollar series cover 1,994 of 2,174 green deals. **Taxonomy anachronism** is a real risk: a 2013 snapshot's categories reflect 2013 ideas about what is green, and sub-domains that matter now may be invisible.

The audit is underdetermined, not reassuring. Emerging-market and advanced-economy false-positive rates are 22.2% and 15.0% with $p = 0.409$. That is an absence of power, not a demonstration of equality, and the headline is therefore reported on raw classifier output, which is the symmetric choice. **H5 has no causal identification**; the depth correlation is descriptive. **Multiple comparisons** are handled by Benjamini–Hochberg across the full family of 28 tests (Benjamini and Hochberg 1995), of which 9 remain significant after adjustment; the full table is Appendix D.

10. Conclusion

Three findings. First, emerging markets hold 15.3–22.7% of world GDP and 1.7–6.4% of the open venture record, and conditional on being in that record an emerging-market company is half as likely to be green as an advanced-economy one — 2.60% against 5.43%, a risk ratio of 0.479 [0.343, 0.669], stable across every green definition and bloc scheme tested. Second, that gap is not green-specific: emerging-market representation falls roughly in proportion to a sector's capital intensity, with an elasticity of -1.19 [-1.64, -0.75], and cleantech sits above rather than below what that fit predicts. Third, the structural questions the field asks about emerging-market green capital cannot be answered from open data, and the six routes by which they fail are independent of one another.

The implication is that the binding constraint on this field is measurement rather than analysis. The concrete next step is narrow and actionable: an instrument field that distinguishes project finance from venture equity. That single change would make the instrument-mix question testable, restore the venture-stage criterion this project had to abandon, and cost nothing to collect at the point the record is created.

Appendix A. Green lexicon vetoes

Tier C exclusion patterns, with the number of companies each removed and the reason it exists. Patterns are regular expressions over the concatenated taxonomy, tag and description fields. The full lexicon, including the 135 Tier B patterns and the ratified manual-exclusion decisions, is `config/green_lexicon.yaml`.

Pattern	Companies	Funded	Reason
\bcleaner(?:s)?\b	127	3	Cleaning products or staff, not clean technology.
\bgreeting(?:s)?\b	65	15	Greeting cards. Substring collision with 'green'.

\b(?:dry[-\s]?clean\w*|clean\w*\s*service(?:s)?|house\s*clean\w*|office\s*clean\w*|janitorial|laundry)\b	48	5	Cleaning and laundry services. 'clean' in a services sense, not an energy sense.
\b(?:oil\s*(?:and|&)?\s*gas|petroleum|petrochemical\w*|upstream\s*oil|oilfield|drilling|fracking|hydraulic\s*fracturing|shale\s*(?:oil|gas)|coal\s*(?:mining|fired|power)|tar\s*sands|oil\s*sands|natural\s*gas\s*(?:production|exploration)|refiner(?:y|ies)\s*services|lng\s*terminal)\b	44	12	Fossil-fuel extraction, refining and coal power. Necessary because the general tier admits a standalone 'Energy' taxonomy token and Crunchbase's 'Energy' bucket mixes renewables with oil, gas and coal. Without this veto an upstream petroleum company classifies as green. Note the deliberate asymmetry: 'biorefinery' and 'hydrogen' remain green, 'refinery services' and 'natural gas production' do not.
\bgreenhouse\b(?!\s*gas)	18	5	Greenhouse Software (recruiting ATS) and ordinary horticulture. Only 'greenhouse gas' is a green signal.
\b(?:solaris|solarium|solariums)\b	8	1	Solaris (operating system) and tanning salons. Word boundaries already prevent the match; logged for completeness.
\b(?:conference(?:s)?|trade\s*show|media\s*compan\w*|magazine|publisher|publishing|news\s*(?:site|portal|outlet))\b.*\b(?:cleantech|greentech|renewable)\b	2	0	Sector media and events businesses. Their capital is not green-venture capital.

Appendix B. Country recovery

Recovery used the flat mirror's country field first and the company homepage's country-code top-level domain second, under an explicit allow-list of 29 domains. 23 domains are refused because they are sold as vanity domains worldwide; accepting them would have manufactured 651 company attributions, concentrated in exactly the countries this paper reports as absent. The accepted and refused lists with per-domain reasons are `config/blocs.yaml`.

The counts below are stated at the scale each is measured on. Company scale covers every company in the snapshot, including the large majority that never raised a recorded round; deal scale covers the 41,105 in-window deals this paper analyses.

Quantity	Scale	Count
Companies in the snapshot	company	196,553
Carrying no country code	company	108,563
Country recovered	company	5,743 (5.3%)
In-window deals	deal	41,105
Country recovered	deal	354
Still unattributable	deal	1,526
— of which no country code	deal	1,520
— of which an unmappable code	deal	6

Appendix C. Robustness

The headline risk ratio under four green definitions.

Specification	EM green	EM %	AE green	AE %	RR	CI low	CI high	p
headline (raw: A or B)	34	2.6	2,056	5.43	0.479	0.343	0.669	7.5×10^{-6}
strict (Tier A only)	29	2.22	1,556	4.11	0.54	0.375	0.776	0.0006
curated (ratified exclusions)	28	2.14	2,030	5.36	0.399	0.276	0.577	2.9×10^{-7}
bare-energy pattern removed	34	2.6	2,035	5.37	0.484	0.346	0.676	1×10^{-5}

The same result under alternative bloc schemes, including a static present-day classification.

Scheme	EM deals	EM %	AE deals	AE %	RR	CI low	CI high	p
time-varying (headline)	1,309	2.6	37,896	5.43	0.479	0.343	0.669	7.5×10^{-6}
static present-day income	1,064	2.82	38,138	5.4	0.522	0.366	0.745	0.0002
MSCI Emerging Markets list	1,489	3.49	38,096	5.4	0.647	0.493	0.847	0.0013

The three curation specifications. Raw applies no curation; curated applies the ratified manual exclusions, which came from an asymmetric audit; bias-corrected applies each bloc's own measured false-positive rate to its whole green population, so neither bloc is penalised for having been audited harder.

Specification	EM %	AE %	RR	95% CI
raw (headline)	2.6	5.43	0.479	[0.3427, 0.6689]
bias-corrected	1.99	4.58	0.433	[0.2576, 0.6407]
partially curated	2.14	5.36	0.399	[0.2761, 0.5774]

Endpoint sensitivity for the within-stage cheque-size result of §6.1. Every admissible start-end pair, so the reader can see which choices reverse the sign rather than taking the headline pair on trust.

Start	End	Span	Green %	Non-green %	Difference pp
2002	2006	4	127.6	−0.3	127.8
2002	2007	5	−84.8	−7.1	−77.8
2002	2008	6	−42.8	−7.8	−35
2002	2009	7	−54.5	−29.5	−25
2002	2010	8	−48.9	−22	−26.9
2002	2011	9	−63.2	−6.4	−56.7
2002	2012	10	−78.4	−8.5	−69.9
2006	2009	3	−6.1	−29	22.9
2006	2010	4	3.6	−21.6	25.1
2006	2011	5	7	−5.5	12.5
2006	2012	6	−28.9	−7.4	−21.5
2007	2010	3	−26.7	−16.3	−10.4
2007	2011	4	−25.5	1	−26.5
2007	2012	5	−50.4	−1.1	−49.3
2008	2011	3	6	1.3	4.7
2008	2012	4	−29.2	−0.8	−28.4

2009	2012	3	−21.7	32.4	−54.1

Appendix D. Test family and multiplicity correction

Every test run in this project, with raw and Benjamini-Hochberg adjusted p-values and an effect size. 9 of 28 survive correction. Tests that could not be run because a cell fell below the minimum size are not listed here; they are reported as unresolved in §6.

H	Test	p raw	Effect	p adjusted	Survives BH
H3	non-dilutive share, green vs non-green (AE)	1.2×10^{-7}	RR 1.4331	3.5×10^{-6}	True
H1	EM vs AE green rate, Fisher exact	1.3×10^{-6}	RR 0.4788	1.8×10^{-5}	True
H3	instrument mix chi-squared (AE)	3.1×10^{-6}	Cramer's V 0.0273	2.9×10^{-5}	True
H1	upper bound, unassigned as EM	6.3×10^{-6}	RR 0.6372	4.2×10^{-5}	True
H1	EM vs AE green rate, two-proportion z	7.5×10^{-6}	RR 0.4788	4.2×10^{-5}	True
H1b	capital-intensity elasticity	1.5×10^{-5}	slope -1.1949	6.8×10^{-5}	True
H5	Spearman green_usd_per_capita vs gdp_per_capita_usd	0.0006	rho 0.6736	0.0024	True
H2	pre-trend differential	0.0012	trend 0.2542/yr	0.0041	True
H5	OLS need after income: electricity_deficit_pct	0.0104	coef 0.222751	0.0325	True
H5	Spearman green_usd_per_gdp vs gdp_per_capita_usd	0.0519	rho 0.4195	0.1448	False
H5	Spearman green_usd_per_capita vs co2_per_capita	0.0665	rho 0.3981	0.1448	False
H5	Spearman green_usd_per_capita vs electricity_deficit_pct	0.0672	rho -0.3972	0.1448	False
H5	Spearman green_usd_per_capita vs electricity_access_pct	0.0672	rho 0.3972	0.1448	False
H5	OLS need after income: co2_per_capita	0.1837	coef 0.116147	0.3675	False
H5	Spearman green_usd_per_gdp vs co2_per_capita	0.2399	rho 0.2614	0.4478	False
Audit	EM vs AE false-positive rate	0.4093	RR 1.4815	0.7164	False
H4	graduation logit, seed to A	0.4583	OR 1.3	0.7549	False
H5	Spearman green_share_of_deals vs renewable_energy_share_pct	0.4934	rho 0.1542	0.7675	False
H5	Spearman green_usd_per_capita vs renewable_energy_share_pct	0.5732	rho -0.127	0.8447	False
H2	DiD green x post	0.658	coef 0.41	0.9211	False
H5	OLS need after income: renewable_energy_share_pct	0.7508	coef 0.005517	0.9821	False
H5	Spearman green_usd_per_gdp vs renewable_energy_share_pct	0.8242	rho -0.0503	0.9821	False
H5	Spearman green_share_of_deals vs electricity_access_pct	0.8532	rho -0.0419	0.9821	False
H5	Spearman green_share_of_deals vs electricity_deficit_pct	0.8532	rho 0.0419	0.9821	False
H5	Spearman green_share_of_deals vs gdp_per_capita_usd	0.9265	rho 0.0209	0.9821	False
H5	Spearman green_usd_per_gdp vs electricity_access_pct	0.9721	rho -0.0079	0.9821	False
H5	Spearman green_usd_per_gdp vs electricity_deficit_pct	0.9721	rho 0.0079	0.9821	False
H5	Spearman green_share_of_deals vs co2_per_capita	0.9821	rho 0.0051	0.9821	False

Appendix E. Attrition ledger

Every filter applied between ingestion and the canonical deal table, with its own before and after counts. The ledger is complete by construction: each filter is applied through a ledger object that records its own effect, so a step cannot be omitted from this table without being omitted from the pipeline.

#	Stage	Action	Reason	In	Out	Dropped
1	ingest	funding_rounds.csv as ingested	starting population	52,928	52,928	0

2	dates	drop unparseable funded_at	no usable deal date, so the row cannot be placed in the window or deflated	52,928	52,680	248
3	dates	drop implausible funded_at	date outside 1990-2014, impossible for a 2013 snapshot	52,680	52,661	19
4	linkage	drop orphan object_id	no matching company in objects.csv, so no country, date of founding or green label	52,661	52,363	298
5	dedupe	drop duplicates on object_id+funded_at+instrument+raised_amount_usd	identical company, date, instrument and amount	52,363	52,259	104
6	join	attach company attributes	left join, must not change row count	52,259	52,259	0
7	window	truncate to 2000-2012	coverage growth contaminates the final year, and the snapshot cut censors December 2013	52,259	41,105	11,154

Appendix F. Leave-one-out sensitivity

Each of the 27 emerging-market green companies removed in turn, with the effect on the emerging-market green rate and on green dollars. The dollar series moves by −20.3% to 0.0% across the set, which is why dollar-denominated emerging-market claims are not made in this report.

Company removed	Country	Deals	Green rate %	Δ pp	Δ USD %
Vital Renewable Energy Company	BRA	1	2.523	−0.075	−20.3
Brenco	BRA	1	2.523	−0.075	−16.1
Isagen	COL	1	2.523	−0.075	−11
Greenko Group	IND	1	2.523	−0.075	−9.1
Quippo Infrastructure	IND	1	2.523	−0.075	−8.9
KSK Power Venture	IND	1	2.523	−0.075	−7.7
Moser Baer Solar	IND	1	2.523	−0.075	−7.3
Akenerji Elektrik Uretim	TUR	1	2.523	−0.075	−6.5
Nitol Solar	RUS	1	2.523	−0.075	−3.9
SunBorne Energy	IND	4	2.299	−0.299	−3.3
Auro Mira Energy	IND	1	2.523	−0.075	−1.6
Red Ambiental	MEX	1	2.523	−0.075	−1.6
Azure Power	IND	3	2.374	−0.224	−1
Attero	IND	2	2.448	−0.149	−0.7
Kotak Urja	IND	1	2.523	−0.075	−0.6
GATe Technology	VNM	1	2.523	−0.075	−0.3
Roshini International Bio Energy	IND	1	2.523	−0.075	−0.1
Villgro Innovation Marketing	IND	2	2.448	−0.149	0
Scion Global	MYS	1	2.523	−0.075	0
Wifinity Technology	IND	1	2.523	−0.075	0
Aventones	CHL	1	2.523	−0.075	0
farmhopping	BGR	1	2.523	−0.075	0
Playground Energy	BGR	1	2.523	−0.075	0
Stima Systems	KEN	1	2.523	−0.075	0
WindGen Power Products	KEN	1	2.523	−0.075	0
SEDEMAC Mechatronics	IND	1	2.523	−0.075	0
Keclon	ARG	1	2.523	−0.075	0

Appendix G. Instrument field disagreements

The project normalises instruments from `funding_round_code`, not from `funding_round_type`. The table below is why: every row is a case where the two fields disagree, with the number of deals affected. It is the non-zero part of the full type-by-code contingency table, which is `outputs/tables/appendix_instrument_type_x_code.csv`; the zero cells are omitted because they carry nothing.

The two fields disagree on 16,419 deals, 31.0% of the source. Two rows carry the argument. `funding_round_type` files 10,189 seed rounds under `angel`, so reading instruments from that field would have collapsed seed into angel and made the graduation funnel in §6.4 unmeasurable. It files 3,409 debt and 776 grant rounds under `other`, so the non-dilutive share that H3 turns on would have been invisible. Neither collapse raises an error; both would have silently changed a result.

Code (used)	Type (discarded)	Deals
seed	angel	10,189
debt_financing	other	3,409
seed	series_a	1,419
grant	other	776
angel	series_a	413
convertible_note	angel	146
convertible_note	series_a	41
secondary_market	other	16
post_ipo_debt	post_ipo_equity	7
crowdfunding	angel	2
crowdfunding	series_a	1

Appendix H. Mirror vintage comparison

The two Crunchbase mirrors are different vintages. This table bounds what that costs: the flat file records funding for companies the relational dump does not, implying under-recording of roughly 6.5% of deals.

Metric	Value
relational max funded_at	2013-12-12
flat max last_funding_at	2015-01-01
flat-only companies	18593
flat-only absent from objects.csv	14511
flat-only in objects.csv but no deal rows	4082
of those, last funded before snapshot close	2186
claimed funding of that gap (USD bn)	14.31
matched companies	30845
matched: flat reports more rounds	6881
matched: equal round counts	23620

matched: flat reports fewer	344
surplus rounds claimed by flat	9765
surplus explained by post-snapshot activity	6353
surplus NOT explained by timing	3412

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